SUPPLY CHAIN DELIVERY PERFORMANCE REPORT

End-to-End Business Insights, Bottleneck Diagnostics & Strategic Recommendations

Period: 2015–2018 | Portfolio Project

1. Executive Summary & Business Context

This report provides an in-depth analysis of global supply chain delivery performance across **180,000+ orders** (2015–2018) for a multi-category e-commerce platform. The company sells sporting goods, fitness equipment, apparel, and footwear across international regions. The core operational challenge identified is a severe gap between scheduled delivery windows and actual shipping arrival times, causing customer distrust, eroded profit margins, and elevated logistics costs.

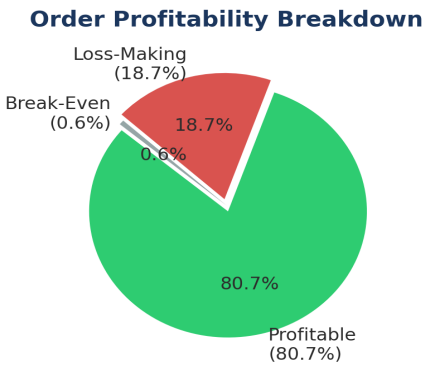
TOTAL ORDERS	LATE DELIVERY RATE	PROFIT AT RISK	ML PREDICTIVE ACC.
180,000+	54.71%	$2.1 Million	74.00%

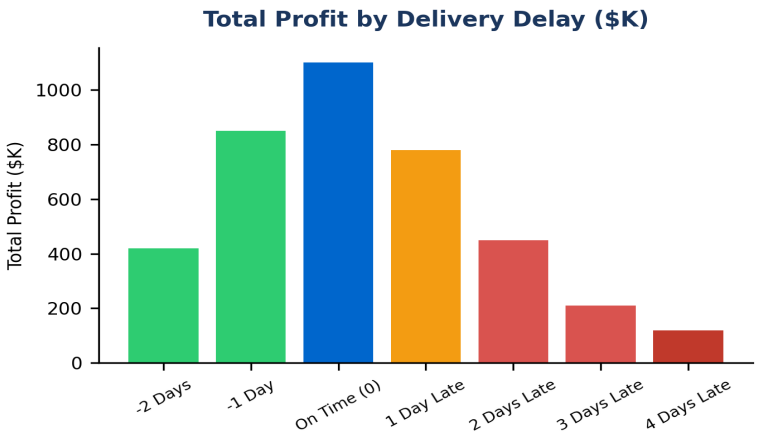
2. Key Performance Indicators (KPI Dashboard)

Metric Name	Current Value	Benchmark / Status	Strategic Impact
Total Orders Analyzed	180,000+	Master Dataset	Baseline scope covering 2015–2018 operations.
On-Time Delivery Rate	45.29%	Critical Deficit (<85%)	Majority of customers experience delayed shipments.
Late Delivery Rate	54.71%	High Friction (>15%)	Core driver of brand erosion and customer churn.
Profit at Risk (Delays)	$2,100,000	Severe Financial Erosion	Eroded profit margins directly linked to delayed orders.
90th Percentile Delay	3 Days	Systemic Containment	Delays are process-driven rather than catastrophic outliers.
Average Order Profit	$22.00	Stable Target ($20–$23)	Unit economics remain profitable across categories.

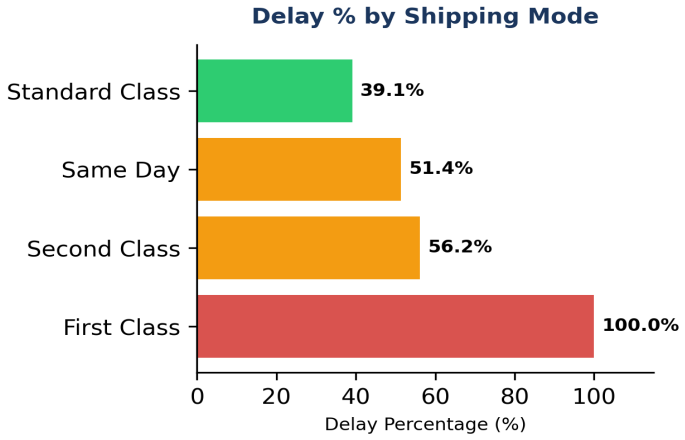
3. Profitability vs. Delivery Delay Analysis

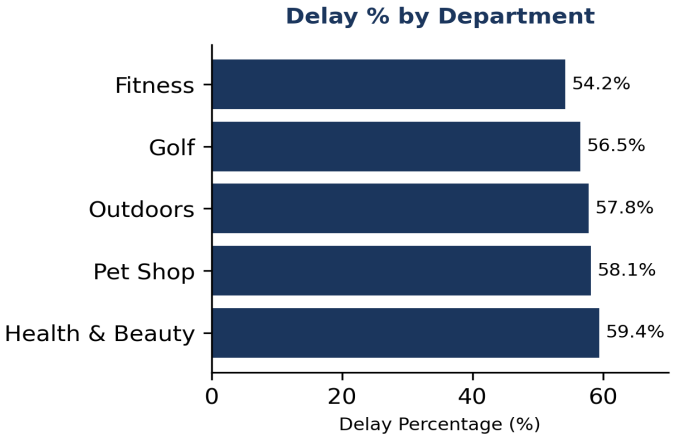
Orders were categorized into three tiers based on unit profit margin: **80.7% Profitable**, **18.7% Loss-Making**, and **0.6% Break-Even**. While the average profit per order remains steady between $20 and $23, delayed orders account for **$2.1M in endangered profit** due to customer concessions, expedited carrier adjustments, and support overhead.





4. Operational Bottleneck Detection & Root Cause Analysis



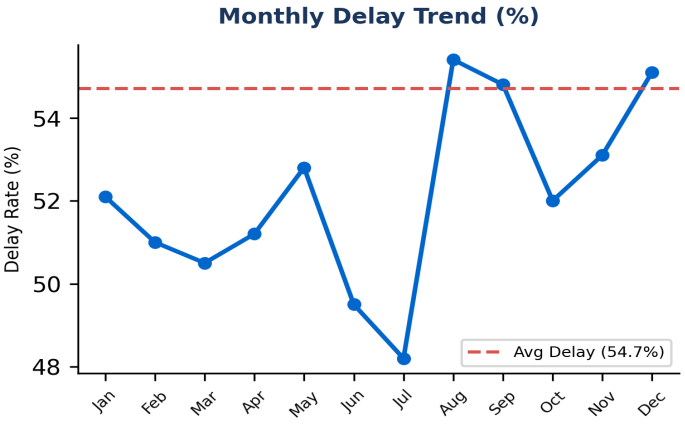


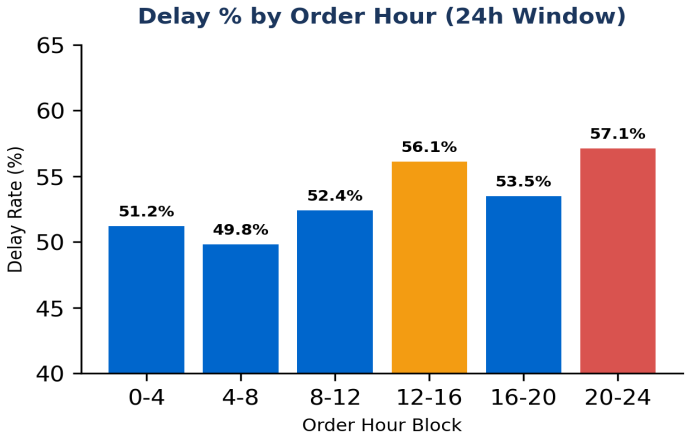
Critical Bottleneck Diagnostic

Key Finding: First Class Shipping has a 100% delay rate! Customers paying premium shipping rates experience guaranteed delivery failures. Across regions, Central Africa leads with 58.7% delays, though all regions cluster narrowly between 55% and 59%, proving a **company-wide systemic failure** rather than localized regional breakdown.

Rank	Top Driver / Factor	Factor Category	Operational Impact
1	**First Class Shipping Mode**	Shipping Mode	100.0% Delay Rate — Complete premium promise breakdown.
2	**Second Class Shipping Mode**	Shipping Mode	56.2% Delay Rate — Heavy carrier congestion.
3	**Payment Review Status**	Order Status	High delay impact due to manual validation holds.
4	**Pending Payment Status**	Order Status	Payment gateway friction halts order dispatch.
5	**Transfer Payment Method**	Payment Type	Wire transfer processing lag stalls fulfillment.
6	**Outdoors & Golf Depts**	Product Department	High product size/handling friction.

5. Time-Based Delay Analysis





• **Seasonal Peak:** August (55.4%), September (54.8%), and December (55.1%) suffer the highest delay rates, driven by back-to-school and holiday order surges.
• **Day of Week:** Orders placed on **Monday and Sunday** exhibit elevated delays due to weekend order backlogs.
• **Hour Block:** Late evening orders placed around **8:00 PM (20:00) reach 57.1% delay rate**, reflecting overnight dispatch queue bottlenecks.

6. Predictive Machine Learning Model Results

To proactively address late deliveries, a **Random Forest Classifier** was trained on order-level features (Scheduled Days, Category, Customer Segment, Department, Order Region, Shipping Mode, Month, Hour). Class imbalance was mitigated using **SMOTE oversampling**.

Evaluation Metric	Model Score	Operational Context
Accuracy	74.00%	Correctly classifies 3 out of 4 orders prior to shipment.
Precision	78.00%	Low false alarm rate when flagging an order as high-risk.
Recall	75.00%	Captures 75% of all potential late orders before dispatch.
F1-Score	76.45%	Harmonic balance between precision and recall reliability.

7. Actionable Business Recommendations

1. Immediate Carrier Audit for Premium Shipping

Audit First Class and Second Class carrier SLAs. Re-negotiate contracts or disable First Class checkout until 100% delay failure is corrected.

2. Real-Time ML Early Warning Alert System

Deploy the Random Forest model into the fulfillment engine to flag high-risk orders (>70% risk score) at checkout for priority dispatch.

3. Automated Payment Verification Escalation

Establish automated alerts for orders stuck in 'Payment Review' or 'Pending Payment' for over 2 hours to clear fulfillment holds.

4. Dynamic Seasonal Surge Capacity Planning

Increase warehouse staffing and carrier allocation by 25% during peak delay months (August, September, December).

5. Default to Standard Class for Eligible Orders

Re-route non-urgent orders to Standard Class (39.1% delay rate) to optimize on-time delivery metrics and logistics costs.

6. Targeted Department Audits

Conduct deep-dive logistics audits for high-delay departments (Health & Beauty, Outdoors, Golf) to address inventory placement friction.

7. Model Retraining & Feature Expansion

Retrain the ML model quarterly; integrate weather, warehouse utilization, and carrier-specific tracking to push target recall above 80%.

8. Conclusion: Current vs. Target State Matrix

Dimension	Current Operational State	Target Operational State	Expected Business Benefit
Delivery SLA	54.71% Late Deliveries	< 15.00% Late Deliveries	Restored customer trust & retention
First Class Delivery	100.0% Failure Rate	> 95.0% On-Time Fulfillment	Elimination of premium refund penalties
Profit Impact	$2.1M Profit at Risk	< $300K Endangered Profit	$1.8M direct margin recovery
Risk Management	Reactive Customer Care	Proactive ML Early Alerts	Automated high-risk order dispatch
Payment Processing	Manual Hold Bottlenecks	< 2hr Auto-Escalation SLA	Elimination of payment-induced delays